

POOJAN DESAI

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TECHNICAL SKILL

Tools: Allen Bradley Studio 5000, Siemens Step 7, FactoryTalk, Siemens TIA, ROS, Gazebo, Open CV, Roboguide, profinet, Modbus, EtherCAT, PlatformIO, RS Links, Wonderware, LabVIEW, MATLAB, Codesys, Ansys, SolidWorks, Fusion 360, AutoCAD, AutoCAD Electrical

Programming Language: Python, C, Ladder Logic, Functional Block Diagram, Structured text.

PROFESSIONAL EXPERIENCE

Kick Robotics (Maryland, USA) - Robotic Engineer

February 2024 - Present

- Designed **CAD models** for Warehouse Bots using **SolidWorks** and **Fusion 360**, incorporating protective covers and electrical systems.
- Conducted **simulations** and **3D printing** for critical components, ensuring precise assembly, achieving **higher safety factor**.
- Participated in optimizing autonomous navigation for warehouse bot using **ROS2** and **Python**, integrating **SLAM, NAV2**, and **sensors** to optimize mapping, obstacle avoidance.
- Worked with the **FANUC** robotic arm, gaining hands-on experience in **RoboGuide** and fundamental robotic arm operations.
- Designed and optimized battery pack enclosures solutions for autonomous warehouse bots, using **SimScale** for simulations for in-house battery production.
- Enhanced **technical documentation**, refining manuals and guides for improved clarity and accessibility.

Naik and Associates (Gujarat, India) - Jr. Safety Engineer

April 2022 - November 2022

- Spearheaded the integration of Industrial Fire and Safety Management Systems, enhancing safety standards and compliance across the Thermal, Oil and Gas, and Chemical industries.
- Facilitated the management of **critical documentation** essential for licence renewal. Additionally, played a key role in **data collection** and **observation** during site visits.
- Fostered strong client relationships by managing comprehensive records for routine audits. Led client meetings, maintaining organized logs of reports and meeting minutes, ensuring professionalism and compliance with industry standards.

Space Technology and Aeronautical Rocketry (Gujarat, India) - Research Engineer

November 2021 - April 2022

- Optimized injector performance using **CFD** simulations, achieving a **10% efficiency** increase, and designed the complete test setup in **Fusion 360 & SolidWorks** for initial testing.
- Managed equipment procurement for the oxidizer tank manufacturing unit, ensuring precision resources for testing.
- Generated detailed research and **testing reports** to document findings and supported data-driven decision-making. Organized client meetings to facilitate seamless communication and foster teamwork in a dynamic engineering environment.

Navsari Agricultural University (Gujarat, India) - Automation Engineer

August 2019 - May 2021

- Automated the patented Banana Pseudostem Process using **SCADA** as the central control system, integrating **PLCs** and **microcontrollers** for efficient operations.
- Designed and implemented control systems with **Siemens S7-1200 PLC** using **TIA Portal** and **STEP 7** for logic development, enhancing precision and efficiency.
- Utilized **Indusoft WebStudio**, **Codesys**, and **Matlab** for process optimization, increasing unit **efficiency by 30%**. Integrated and calibrated **VFDs** and **sensors** (level, flow, temperature) for real-time monitoring and improved system control.
- Developed **P&IDs** and managed end-to-end automation workflows, including **HMI setup**, SCADA configuration, and troubleshooting.
- Led **FAT** and **SAT** activities to ensure seamless performance during installation and operation.

PROFESSIONAL PROJECTS

Automatic Drink Dispenser Unit

- Developed an automated drink dispenser using **Allen-Bradley CompactLogix 5480 PLC** and **Studio 5000**, implementing robust ladder logic for precise dispensing control.
- Integrated flow meters and level sensors for real-time fluid monitoring, ensuring accurate measurement and automated pump and valve control.
- Implemented **Modbus** protocol for seamless data exchange between the PLC and peripheral devices, optimizing hardware coordination.
- Programmed and calibrated the system to manage multiple drink recipes, enhancing customization, reliability, and efficiency with diagnostics and troubleshooting.

Water Treatment and Monitoring System

- Designed and implemented an automated system for real-time monitoring and control of a simulated water treatment process, integrating **Allen-Bradley CompactLogix PLC** and **FactoryTalk View HMI**.
- Configured **sensor** for flow, pH, and temperature using **Modbus** communication, ensuring accurate data acquisition and automated process adjustments.
- Programmed and optimized **Ladder logic** for pump operations, and filtration processes, enhancing efficiency and minimizing wastage.
- Conducted system troubleshooting and performance analysis, ensuring seamless data logging, alarm management, and capabilities for effective process control.

Real Time Hand Movement Tracking and Mimicking Robot

- Collaborated with a cross-functional team of five members in the development of a Real-Time Hand Movement Tracking and Mimicking Robot, achieving successful integration of a Mediapipe model into **ROS** and **Gazebo**.
- Developed the hand model using **Solidworks** and the created the **URDF** of the model.
- Compiled a comprehensive project report, offering detailed insights into the development process and outcomes of the Real-Time Hand Movement Tracking and Mimicking Robot.

EDUCATION

Master of Engineering in Robotics, University of Maryland, College Park	2023 - 2024
• Course work includes: Robot Modeling and Controls, Aerial Robotics, Manufacturing and Automation	GPA: 3.8/4
Bachelor of Engineering in Instrumentation and Control, Gujarat Technological University, India	2017 - 2021
• Course work includes: PLC, DCS/SCADA, Control Systems, Industrial Instruments	GPA: 3.65/4